TAFS API Documentation

Bill Inquiry & Bill Payment Integration
for United Bank Limited (UBL)

The American Foundation School (TAFS)
TAFS Development Team

Draft v0.1 – 11 October 2026

Purpose of This Document

This document describes the interface TAFS provides to United Bank Limited (UBL) for collecting school fees through UBL channels. It covers:

- the collection channels and how a payer identifies a student on each;
- the **Bill Inquiry** (`GET`) and **Bill Payment** (`POST`) APIs, with request and response samples for success and failure;
- the amount rules: within due date, after due date, and the 1Bill aggregator fee;
- mandatory tracking of the **payment channel** and **payment method** on every payment;
- TAFS's stated consent to resolving every search by CC to the student's most recent voucher;
- the points on which TAFS asks UBL to confirm feasibility (Section 9).

Every payload in this document matches what the TAFS implementation sends and accepts.

1 Collection Channels

Channel	How the payer pays	Identifier entered
UBL branch (OTC)	Cash, cheque or pay order at any UBL branch counter.	The student's **CC number**.
UBL digital app	Bill Payment → select the **TAFS** beneficiary.	The student's **CC number**.
UBL internet banking	Bill Payment → select the **TAFS** beneficiary.	The student's **CC number**.
1Bill (any bank / wallet)	"1Bill Invoices" option in any participating bank or wallet.	The **1Bill consumer number** (Section 2.2).

These instructions are printed at the bottom of every TAFS fee voucher issued for collection into the UBL account.

2 Identifying a Student

2.1 CC number

Every TAFS student has a **CC number**: a numeric student code of up to 4 digits (e.g. `8088`). The CC is printed on every voucher. It is the identifier for OTC, the UBL app and internet banking. Leading zeros may be sent or omitted (`0044` and `44` identify the same student).

2.2 1Bill consumer number

For 1Bill, the consumer number is a fixed 14-digit number:

`[10-digit TAFS 1Bill prefix][CC as 4 digits, zero-padded]`

Example: with prefix `1000000000` and CC `8088`, the 1Bill consumer number is `10000000008088`; for CC `44` it is `10000000000044`.

Request to UBL: 10-digit 1Bill prefix. TAFS requests UBL to issue (or arrange through 1Bill) the **10-digit 1Bill prefix** for The American Foundation School and share it with TAFS. Until it is received, TAFS vouchers and this document use the placeholder `1000000000`. Once UBL shares the prefix, TAFS will switch to it by configuration; no code change is needed on either side.

2.3 Voucher number

Vouchers collected through UBL carry a plain serial voucher number (e.g. `12461`), printed small at the top of the voucher. UBL does **not** need it to look up a bill. It is returned by the inquiry for reference, and UBL may echo it in the payment request (Section 7).

3 Voucher Resolution: Latest Voucher Rule & TAFS Consent

Every inquiry and payment made with a CC or a 1Bill consumer number resolves to **the most recent voucher generated by the school for that student**. Concretely, it is the newest voucher, excluding:

- vouchers the school has cancelled because a newer voucher replaced them; and
- vouchers the school has generated but not yet released to the parent.

If that voucher is already paid, has expired, or has been written off, the API says so and no older voucher is offered in its place.

Statement of consent. The American Foundation School confirms that it has requested this method and consents to it: every search by CC number (or by 1Bill consumer number), on every UBL channel, returns the most recent voucher generated by the school for that student as defined above. TAFS is satisfied that this is the correct bill to present and collect. TAFS accepts that UBL bears no responsibility for a payment being applied to that voucher rather than to any earlier voucher of the same student. Every voucher generated by the school carries the full amount the student owes, including any arrears.

4 Amount Rules

Situation	Amount payable
On or before the due date	The voucher amount (`amount_within_due_date`).
After the due date (until validity date)	Voucher amount + late payment surcharge, normally **PKR 1,000** (`amount_after_due_date`).
After the validity date	Not payable. The school must issue a new voucher (response code `04`).
Channel = `ONEBILL`	The amounts above **+ PKR 30** (1Bill aggregator fee).
OTC, app, internet banking	The amounts above, with no addition.

- The inquiry returns `amount_payable`, the exact amount to collect *today* on the channel stated in the request.

- The payment must carry exactly that amount. Partial payments and overpayments are rejected (code `06`).
- On a 1Bill payment TAFS records only the fee amount against the student. The PKR 30 is recorded as the 1Bill aggregator's fee and is never credited to the student's account.
- Dates are evaluated in Pakistan Standard Time (UTC+05:00).

5 Connectivity & Authentication

5.1 Base URL

```
https://tafs-backend-production.up.railway.app/api/v1/ubl
```

TAFS will also provide a separate test (UAT) base URL with test students before go-live.

5.2 Authentication

Every request must carry two HTTP headers. Credentials are issued by TAFS over a secure channel and are never sent in the URL.

Header	Type	Description
`X-UBL-User`	string	Service user ID issued to UBL
`X-UBL-Password`	string	Password for that service user
`Content-Type`	string	`application/json` (payment requests)

A request with missing or wrong credentials receives response code `10`. All traffic is HTTPS (TLS 1.2 or higher). TAFS can additionally restrict access to UBL's outbound IP addresses on request.

5.3 HTTP status

A well-formed, authenticated request always receives **HTTP 200**. The business outcome is in `response_code`. A malformed request receives **HTTP 400** with a description of the invalid field, for example a missing `channel` or a non-numeric `consumer_number`.

6 API 1 – Bill Inquiry (GET)

```
GET /api/v1/ubl/bill-inquiry?consumer_number={CC or 1Bill no}&channel={channel}
```

6.1 Query parameters

Parameter	Type	Required	Description
`consumer_number`	digits	Yes	CC number (OTC, app, internet banking) or 14-digit 1Bill consumer number.
`channel`	enum	**Yes**	`ONEBILL`, `OTC`, `APP` or `INTERNET_BANKING`. Decides whether PKR 30 is added.

6.2 Successful response fields

Field	Type	Description
response_code	string	"00"
response_desc	string	"Success"
consumer_number	string	As sent
onebill_consumer_number	string	The student's 14-digit 1Bill number
cc	string	Student CC number
student_name	string	Student's full name
gr_number	string	School registration number
campus	string	Campus name
class	string	Class
voucher_number	string	Voucher being collected (plain serial)
billing_month	string	Month(s) billed, e.g. "OCT 26"
voucher_status	string	UNPAID, OVERDUE or PARTIALLY_PAID
issue_date	string	YYYYMMDD
due_date	string	YYYYMMDD
validity_date	string	YYYYMMDD; not payable after this date
currency	string	"PKR"
amount_within_due_date	number	Payable on or before the due date (incl. channel fee)
amount_after_due_date	number	Payable after the due date (incl. late fee and channel fee)
late_fee	number	Late payment surcharge applicable after the due date
channel	string	As sent
channel_fee	number	30 for ONEBILL, else 0
amount_payable	number	**Exact amount to collect today**

6.3 Sample – 1Bill inquiry before the due date

```
GET /api/v1/ubl/bill-inquiry?consumer_number=10000000008088&channel=ONEBILL
X-UBL-User: <issued by TAFS>
X-UBL-Password: <issued by TAFS>
```


```
{
  "response_code": "00",
  "response_desc": "Success",
  "consumer_number": "10000000008088",
  "onebill_consumer_number": "10000000008088",
  "cc": "8088",
  "student_name": "HAFSA DAHRI",
  "gr_number": "7001",
  "campus": "Gulistan-e-Jauhar Campus",
  "class": "AS",
  "voucher_number": "12461",
  "billing_month": "OCT 26",
  "voucher_status": "UNPAID",
  "issue_date": "20261001",
  "due_date": "20261010",
  "validity_date": "20261020",
  "currency": "PKR",
  "amount_within_due_date": 90030,
  "amount_after_due_date": 91030,
  "late_fee": 1000,
```

```
  "channel": "ONEBILL",
  "channel_fee": 30,
  "amount_payable": 90030
}
```

6.4 Sample – OTC inquiry after the due date

```
GET /api/v1/ubl/bill-inquiry?consumer_number=8088&channel=OTC
```

```
{
  "response_code": "00",
  "response_desc": "Success",
  "cc": "8088",
  "voucher_number": "12461",
  "voucher_status": "OVERDUE",
  "amount_within_due_date": 90000,
  "amount_after_due_date": 91000,
  "late_fee": 1000,
  "channel": "OTC",
  "channel_fee": 0,
  "amount_payable": 91000,
  ...
}
```

6.5 Sample – failure

A failed inquiry returns only `response_code`, `response_desc` and, where a voucher was found, `voucher_number`:

```
{ "response_code": "03", "response_desc": "Voucher is already paid", "voucher_number
    ": "12461" }
```

7 API 2 – Bill Payment (POST)

```
POST /api/v1/ubl/bill-payment
Content-Type: application/json
```

7.1 Request body

Field	Type	Required	Description
`consumer_number`	digits	Yes	Same value used in the inquiry
`channel`	enum	**Yes**	`ONEBILL`, `OTC`, `APP`, `INTERNET_BANKING`
`payment_method`	enum	**Yes**	How the payer paid. See Section 7.2
`amount`	number	Yes	Amount collected from the payer, PKR. Must equal `amount_payable` (incl. PKR 30 on 1Bill)
`transaction_id`	string	Yes	UBL's unique transaction reference (max 100). Used to prevent duplicate posting

Field	Type	Required	Description
`transaction_datetime`	ISO 8601	Yes	When the payment was taken, with offset, e.g. `2026-10-05T11:00:00+05:00`
`voucher_number`	string	Advised	`voucher_number` from the inquiry; guards against a newer voucher issued in between
`branch_code`	string	OTC	UBL branch code where cash/cheque was received
`payer_bank_code`	string	1Bill	Code of the bank or wallet the payer paid from
`payer_bank_name`	string	1Bill	Name of that bank or wallet
`instrument_number`	string	Cheque / PO	Cheque or pay-order number. Required when `payment_method` is `CHEQUE` or `PAY_ORDER`

7.2 Mandatory payment method tracking

TAFS requires every payment to state **both** the channel it arrived through and the method the payer actually used, so that each receipt can be reconciled. A payment without them is rejected (HTTP 400).

`payment_method`	Meaning
`CASH`	Cash at a branch counter
`CHEQUE`	Cheque (send `instrument_number`)
`PAY_ORDER`	Pay order / demand draft (send `instrument_number`)
`ACCOUNT_TRANSFER`	Debit from the payer's bank account (app, internet banking, 1Bill)
`DEBIT_CARD`	Debit card
`CREDIT_CARD`	Credit card
`WALLET`	Mobile wallet (e.g. via 1Bill)
`OTHER`	Any other method; please tell TAFS which so it can be added

7.3 Successful response

Field	Type	Description
`response_code`	string	`"00"`
`response_desc`	string	`"Success"`
`transaction_id`	string	As sent
`tafs_receipt_id`	string	TAFS receipt number for this payment
`voucher_number`	string	Voucher the payment was applied to
`amount_booked`	number	Amount credited to the student (excludes the 1Bill fee)
`channel_fee`	number	1Bill fee retained by the aggregator (`30` or `0`)

7.4 Sample – 1Bill payment

```
POST /api/v1/ubl/bill-payment
{
  "consumer_number": "10000000008088",
```

```
  "channel": "ONEBILL",
  "payment_method": "ACCOUNT_TRANSFER",
  "amount": 90030,
  "transaction_id": "UBL20261005110000123",
  "transaction_datetime": "2026-10-05T11:00:00+05:00",
  "voucher_number": "12461",
  "payer_bank_code": "HBL",
  "payer_bank_name": "Habib Bank Limited"
}
```

```
{
  "response_code": "00",
  "response_desc": "Success",
  "transaction_id": "UBL20261005110000123",
  "tafs_receipt_id": "48215",
  "voucher_number": "12461",
  "amount_booked": 90000,
  "channel_fee": 30
}
```

7.5 Sample – OTC cheque payment after the due date

```
{
  "consumer_number": "8088",
  "channel": "OTC",
  "payment_method": "CHEQUE",
  "instrument_number": "00412387",
  "branch_code": "0457",
  "amount": 91000,
  "transaction_id": "UBLOTC0457202610121015",
  "transaction_datetime": "2026-10-12T10:15:00+05:00",
  "voucher_number": "12461"
}
```

7.6 Sample – failures

```
{ "response_code": "06", "response_desc": "Amount does not match the payable amount
    ", "amount_payable": 90030 }
{ "response_code": "08", "response_desc": "Duplicate transaction; already posted",
  "transaction_id": "UBL20261005110000123", "tafs_receipt_id": "48215" }
```

7.7 Duplicates, retries and timeouts

- Posting is idempotent on `transaction_id`. If UBL does not receive a response (timeout), it should resend the **same** request with the **same** `transaction_id`. A payment that was already posted returns code `08` with the original `tafs_receipt_id` and is never booked twice.
- Code `08` therefore means the payment **is** recorded at TAFS and should be treated as a success.
- Code `99` means nothing was booked; UBL may retry.

8 Response Codes

Code	Meaning	Used by
00	Success	Inquiry, Payment
01	Consumer number not found	Inquiry, Payment
02	No voucher has been issued for this student	Inquiry, Payment
03	Voucher is already paid	Inquiry, Payment
04	Voucher validity has expired; school must issue a new voucher	Inquiry, Payment
05	Voucher is not payable (written off)	Inquiry, Payment
06	Amount does not match the payable amount	Payment
07	A newer voucher has been issued; inquire again	Payment
08	Duplicate transaction; already posted (treat as success)	Payment
09	Invalid request (e.g. cheque without `instrument_number`)	Payment
10	Invalid credentials	Inquiry, Payment
99	General error; nothing booked, retry later	Inquiry, Payment

9 Points for UBL to Confirm

TAFS asks UBL to confirm the following before integration testing:

1. **1Bill channel on inquiry.** Can UBL tell TAFS, on the `GET` inquiry itself, that the request came through 1Bill (the `channel=ONEBILL` parameter), so that the PKR 30 aggregator fee is included in the amount shown to the payer? If 1Bill inquiries reach TAFS in a different way, for example through a separate 1Bill-facing endpoint or with different credentials, please describe it and TAFS will adapt.

2. **1Bill fee settlement.** Is the PKR 30 collected from the payer on top of the bill and retained by the aggregator, so that UBL settles only the fee amount into the TAFS account? TAFS's books assume this.

3. **Payment method on the payment request.** Can UBL send `payment_method` on every payment, for every channel, using the values in Section 7.2? For 1Bill, can UBL also pass the paying bank or wallet (`payer_bank_code` / `payer_bank_name`)?

4. **OTC search by CC.** Can the OTC teller screen search by CC number and display the inquiry result before accepting payment?

5. **Cheques.** Are cheques posted to TAFS when deposited, or only once cleared? If posted at deposit, TAFS will need a way to be told about a returned cheque (see item 7).

6. **1Bill prefix (requested).** Please issue TAFS's 10-digit 1Bill prefix (Section 2.2) and confirm when it will be live.

7. **Reversals.** This version has no reversal API. A payment posted in error is corrected by the school on request. Please tell TAFS if UBL requires an online reversal message, and its format.

8. **Reconciliation.** Can UBL share a daily settlement statement listing `transaction_id`, amount, channel and payment method, so TAFS can reconcile against its receipts?

9. **Message format.** If UBL requires its own standard message format (field names, fixed-length amounts, response codes), please share the specification and TAFS will map to it.

10 What Is Printed on the Voucher

Each voucher issued for collection into the UBL account carries, at the bottom:

- **UBL branches:** pay at any UBL branch by quoting the student's CC number.
- **UBL digital app:** go to Bill Payment, select "TAFS" as the beneficiary and enter the student's CC number.
- **Other banks / digital channels (1Bill):** pay via "1Bill Invoices" using the 1Bill consumer number printed on the voucher; 1Bill adds a PKR 30 service charge.

For The American Foundation School	For United Bank Limited
Name / Designation / Date	Name / Designation / Date